

## LECTURE 5

### HYPOTHESIS TESTING

### Basic concepts

In this lecture, we continue to discuss the normal classical linear regression model defined by Assumptions (A1)–(A5). Let  $\theta \in \Theta \subset \mathbb{R}^d$  be a parameter of interest. Some examples of  $\theta$  include:

- The coefficient of one of the regressors:  $\theta = \beta_1$ ,  $d = 1$ ,  $\Theta = \mathbb{R}$ .
- A vector of coefficients:  $\theta = (\beta_1, \dots, \beta_l)'$ ,  $d = l$ ,  $\Theta = \mathbb{R}^l$ .
- The variance of errors:  $\theta = \sigma^2$ ,  $d = 1$ ,  $\Theta = \mathbb{R}_{++}$ .

A statistical hypothesis is an assertion about  $\theta$ . Usually, we have two competing hypotheses, and we want to draw a conclusion, based on the data, as to which of the hypotheses is true. Let  $\Theta_0 \subset \Theta$  and  $\Theta_1 \subset \Theta$  such that  $\Theta_0 \cap \Theta_1 = \emptyset$  and  $\Theta_0 \cup \Theta_1 = \Theta$ . The two competing hypotheses are:

- Null hypothesis  $H_0 : \theta \in \Theta_0$ . This is a hypothesis that is held as true, unless the data provide *sufficient* evidence against it.
- Alternative hypothesis  $H_1 : \theta \in \Theta_1$ . This is a hypothesis against which the null is tested. It is held to be true if the null is found false.

The subsets  $\Theta_0$  and  $\Theta_1$  are chosen by the econometrician, and therefore are *known*. Usually, the econometrician has to carry the burden of proof, so the case that they are interested in is stated as  $H_1$ .

The two hypotheses,  $H_0$  and  $H_1$ , must be *disjoint*. Their union defines the *maintained* hypothesis, that is, the space of values that  $\theta$  can take. For example, when  $\Theta = \mathbb{R}$ , one may consider  $\Theta_0 = \{0\}$  and  $\Theta_1 = \mathbb{R} \setminus \{0\}$ . Another example is  $\Theta_0 = (-\infty, 0]$  and  $\Theta_1 = (0, \infty)$ .

When  $\Theta_0$  has exactly one element ( $\Theta_0$  is a singleton), we say that  $H_0 : \theta \in \Theta_0$  is a *simple* hypothesis. Otherwise, we say that  $H_0$  is a *composite* hypothesis. Similarly,  $H_1 : \theta \in \Theta_1$  can be simple or composite depending on whether  $\Theta_1$  is a singleton or not.

The econometrician has to choose between  $H_0$  and  $H_1$ . The *decision rule* that leads the econometrician to *accept* or *reject*  $H_0$  is based on a *test statistic*, which is a function of the data ( $X$  and  $Y$  in the case of a regression model). Let  $S \in \mathcal{S}$  denote a test statistic, where  $\mathcal{S}$  is the range of its values. A decision rule is defined by a partition of  $\mathcal{S}$  into an acceptance region  $\mathcal{A}$  and a rejection (critical) region  $\mathcal{R}$ . The acceptance and rejection regions must be disjoint ( $\mathcal{A} \cap \mathcal{R} = \emptyset$ ), and their union must be equal to the range of possible values for  $S$  ( $\mathcal{A} \cup \mathcal{R} = \mathcal{S}$ ). One rejects  $H_0$  when the test statistic falls into the *rejection region*:  $S \in \mathcal{R}$ . Thus, tests can be described by their decision rules: Reject  $H_0$  when  $S \in \mathcal{R}$ .

There are two types of errors that the econometrician can make:

- Type I error is the error of rejecting  $H_0$  when  $H_0$  is true.
- Type II error is the error of accepting  $H_0$  when  $H_1$  is true.

The probabilities of Type I and II errors can be described using the so-called *power function*. Consider a test based on  $S$  that rejects  $H_0$  when  $S \in \mathcal{R}$ . The power function of this test is defined as:

$$\pi(\theta) = \Pr_{\theta}(S \in \mathcal{R}),$$

where  $\Pr_{\theta}(\cdot)$  denotes the probability calculated under the assumption that the true value of the parameter is  $\theta$ . Thus, the power function of a test gives the probability of rejecting  $H_0$  for every possible value of  $\theta$ . The largest probability of Type I error (rejecting  $H_0$  when it is true) is

$$\sup_{\theta \in \Theta_0} \pi(\theta) = \sup_{\theta \in \Theta_0} \Pr_{\theta}(S \in \mathcal{R}). \tag{1}$$

The expression above is also called the *size* of a test. When  $H_0$  is simple, that is,  $\Theta_0 = \{\theta_0\}$ , the size can be computed simply as  $\pi(\theta_0) = \Pr_{\theta_0}(S \in \mathcal{R})$ .

The probability of Type II error (accepting  $H_0$  when it is false) is:

$$1 - \pi(\theta) = 1 - \Pr_{\theta}(S \in \mathcal{R}) \quad \text{for } \theta \in \Theta_1. \quad (2)$$

Typically,  $\Theta_1$  has many elements, and therefore the probability of Type II error depends on the true value  $\theta$ . One would like the probabilities of both Type I and Type II errors to be as small as possible, but unfortunately, they are inversely related, as is apparent from (1) and (2). To reduce the probability of Type I error (falsely rejecting  $H_0$ ), one should make  $\mathcal{R}$  smaller. This, however, will increase the probability of Type II error.

By convention, a *valid* test must control the size (probability of Type I error). This is consistent with the idea that the econometrician must carry the burden of proof (recall that the econometrician must state their preferred hypothesis as  $H_1$ ).

**Definition.** A test with power function  $\pi(\theta)$  is said to be a level  $\alpha$  test if  $\sup_{\theta \in \Theta_0} \pi(\theta) \leq \alpha$ . We say it is a size  $\alpha$  test if  $\sup_{\theta \in \Theta_0} \pi(\theta) = \alpha$ .

Every size  $\alpha$  test is a level  $\alpha$  test. We consider a test to be valid if it is a level  $\alpha$  test for some pre-chosen  $\alpha \in (0, 1)$ , where  $\alpha$  is called the *significance level* of a test. Typically, the significance level is chosen to be a small number close to zero: for example,  $\alpha = 0.01, 0.05, 0.10$ .

The following are the steps of hypothesis testing:

1. Specify  $H_0$  and  $H_1$ .
2. Choose the significance level  $\alpha$ .
3. Define a decision rule (a test statistic and a rejection region) so that the resulting test is a level  $\alpha$  test.
4. Perform the test.

The decision depends on the significance level: it is easier to reject the null for larger values of  $\alpha$ , since they correspond to larger rejection regions. Given data, the smallest significance level at which the null can be rejected is called the *p-value*. Instead of reporting test outcomes (accept or reject) for some specific  $\alpha$ , it is also common to report *p-values*:

1. Specify  $H_0$  and  $H_1$ .
2. Define a test.
3. Compute the *p-value*.
4.  $H_0$  is rejected for all values of  $\alpha$  that are *greater* than the *p-value*.

The *power* of a test with power function  $\pi(\theta)$  is defined as

$$\pi(\theta) \quad \text{for } \theta \in \Theta_1.$$

Given two level  $\alpha$  tests, we should prefer a more powerful test. We say that a level  $\alpha$  test with power function  $\pi_1(\theta)$  is *uniformly* more powerful than a level  $\alpha$  test with power function  $\pi_2(\theta)$  if  $\pi_1(\theta) \geq \pi_2(\theta)$  for all  $\theta \in \Theta_1$ . As will be apparent from the next section, tests based on estimators with smaller variances typically result in uniformly more powerful tests.

## Testing a hypothesis about a single coefficient

Consider the partitioned regression discussed in Lecture 4:

$$Y = \beta_1 X_1 + X_2 \beta_2 + U,$$

where  $X_1$  is the  $n \times 1$  vector of the observations of the first regressor. Assume that the disturbance variance  $\sigma^2$  is known. Let  $\hat{\beta}_1$  be the LS estimator of  $\beta_1$ . Suppose we want to test

$$\begin{aligned} H_0 : & \beta_1 = \beta_{1,0}, \\ H_1 : & \beta_1 \neq \beta_{1,0}. \end{aligned} \tag{3}$$

Confidence intervals and hypothesis testing are closely related. In fact, a decision rule for an  $\alpha$ -level test can be based on the  $CI_{1-\alpha}$ . The  $1 - \alpha$  level confidence interval for  $\beta_1$  is

$$CI_{1-\alpha} = \left[ \hat{\beta}_1 - z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)}, \hat{\beta}_1 + z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)} \right].$$

Consider the following test:

$$\text{Reject } H_0 \text{ if } \beta_{1,0} \notin CI_{1-\alpha}.$$

The critical region in this case is given by the complement of the  $CI_{1-\alpha}$ . Thus, we reject if

$$\begin{aligned} \beta_{1,0} &< \hat{\beta}_1 - z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)}, \text{ or} \\ \beta_{1,0} &> \hat{\beta}_1 + z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)}. \end{aligned}$$

Equivalently, we reject if

$$\left| \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} \right| > z_{1-\alpha/2}. \tag{4}$$

Such a test is called *two-sided* since, under the alternative, the true value of  $\beta_1$  may be smaller or larger than  $\beta_{1,0}$ .

The expression on the left-hand side is the test statistic. To compute the rejection probability, assume that the true value is  $\beta_1$  and write

$$\frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} = \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} + \frac{\beta_1 - \beta_{1,0}}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}}. \tag{5}$$

Then

$$\frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} \mid X \sim N \left( \frac{\beta_1 - \beta_{1,0}}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}}, 1 \right).$$

If the null hypothesis is true, then  $\beta_1 - \beta_{1,0} = 0$ , and the test statistic has a standard normal distribution. In this case, by the definition of  $z_{1-\alpha/2}$ ,

$$\begin{aligned} \Pr(\text{Reject } H_0 \mid X, H_0 \text{ true}) &= \Pr \left( \left| \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} \right| > z_{1-\alpha/2} \mid X \right) \\ &= \alpha. \end{aligned}$$

Thus, the suggested test has the correct size  $\alpha$ . If the null hypothesis is false, the distribution of the test statistic is not centered around zero, and we will see rejection rates higher than  $\alpha$ .

The probability of rejection is a function of the true value  $\beta_1$  and depends on the magnitude of the second term in (5),  $|\beta_1 - \beta_{1,0}| / \sqrt{\sigma^2 / (X_1^\top M_2 X_1)}$ . For example, suppose that

$$\begin{aligned}\beta_{1,0} &= 0, \\ \sqrt{\sigma^2 / (X_1^\top M_2 X_1)} &= 1, \\ \alpha &= 0.05 \quad (z_{1-\alpha/2} = 1.96).\end{aligned}$$

Let  $Z \sim N(0, 1)$ . In this case, the *power function* of the test is

$$\begin{aligned}\pi(\beta_1) &= \Pr \left( \left| \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} \right| > z_{1-\alpha/2} \mid X \right) \\ &= \Pr \left( \left| \frac{\hat{\beta}_1 - \beta_1 + \beta_1 - \beta_{1,0}}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} \right| > 1.96 \mid X \right) \\ &= \Pr(|Z + \beta_1| > 1.96) \\ &= \Pr(Z < -1.96 - \beta_1) + \Pr(Z > 1.96 - \beta_1).\end{aligned}$$

For example,

$$\pi(\beta_1) = \begin{cases} 0.52 & \text{for } \beta_1 = -2, \\ 0.17 & \text{for } \beta_1 = -1, \\ 0.05 & \text{for } \beta_1 = 0, \\ 0.17 & \text{for } \beta_1 = 1, \\ 0.52 & \text{for } \beta_1 = 2. \end{cases}$$

In this case, the power function is minimized at  $\beta_1 = \beta_{1,0}$ , where  $\pi(\beta_1) = \alpha$ .

These power calculations assume the significance level is known in advance. A closely related question is how to summarize the strength of evidence against the null for a given realization of the test statistic. This is the role of the *p*-value. To illustrate, suppose that, given the data, the test statistic in (4) is equal to 1.88. For the standard normal distribution,  $\Pr(Z > 1.88) = 0.03$ . Therefore, the *p*-value for the two-sided test is  $2 \times 0.03 = 0.06$ . The null is rejected at any significance level exceeding 0.06.

In the case of unknown  $\sigma^2$ , one can test (3) by considering the *t*-statistic:

$$\begin{aligned}T &= \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{s^2 / (X_1^\top M_2 X_1)}} \\ &= \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\widehat{\text{Var}}(\hat{\beta}_1 \mid X)}}.\end{aligned}\tag{6}$$

The test is given by the following decision rule:

$$\text{Reject } H_0 \text{ if } |T| > t_{n-k, 1-\alpha/2}.$$

The *t*-distribution result for the ratio of the OLS estimator to its standard error (Lecture 4) then implies that, under  $H_0$ ,  $\Pr(|T| > t_{n-k, 1-\alpha/2} \mid X, H_0 \text{ true}) = \alpha$ .

One can also consider *one-sided* tests. In the case of one-sided tests, the null and alternative hypotheses may be specified as

$$\begin{aligned}H_0 : & \quad \beta_1 \leq \beta_{1,0}, \\ H_1 : & \quad \beta_1 > \beta_{1,0}.\end{aligned}$$

Both  $H_0$  and  $H_1$  are composite, and the probability of rejection varies not only across the values of  $\beta_1$  specified under  $H_1$  but also across  $H_0$ . In this case, a valid test should satisfy the following condition:

$$\sup_{\beta_1 \leq \beta_{1,0}} \Pr(\text{reject } H_0 \mid X, \beta_1) \leq \alpha, \quad (7)$$

i.e., the maximum probability of rejecting  $H_0$  when it is true should not exceed  $\alpha$ . Let  $T$  be as defined in (6) and consider the following test (decision rule):

$$\text{Reject } H_0 \text{ when } T > t_{n-k, 1-\alpha}.$$

Under  $H_0$ , we have:

$$\begin{aligned} \Pr(\text{reject } H_0 \mid X, \beta_1 \leq \beta_{1,0}) &= \Pr(T > t_{n-k, 1-\alpha} \mid X, \beta_1 \leq \beta_{1,0}) \\ &= \Pr\left(\frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{s^2 / (X_1^\top M_2 X_1)}} > t_{n-k, 1-\alpha} \mid X, \beta_1 \leq \beta_{1,0}\right) \\ &\leq \Pr\left(\frac{\hat{\beta}_1 - \beta_1}{\sqrt{s^2 / (X_1^\top M_2 X_1)}} > t_{n-k, 1-\alpha} \mid X, \beta_1 \leq \beta_{1,0}\right) \quad (\text{since } \beta_1 \leq \beta_{1,0}) \\ &= \alpha \quad (\text{since } \frac{\hat{\beta}_1 - \beta_1}{\sqrt{s^2 / (X_1^\top M_2 X_1)}} \mid X \sim t_{n-k}). \end{aligned}$$

Thus, the size control condition (7) is satisfied. Since this is a one-sided test, the entire rejection probability is assigned to the right tail of the distribution.

## Testing a single linear restriction

Consider the normal linear regression model defined by Assumptions (A1)–(A5):

$$Y = X\beta + U.$$

Suppose we want to test

$$\begin{aligned} H_0 : & \quad c^\top \beta = r, \\ H_1 : & \quad c^\top \beta \neq r. \end{aligned}$$

In this case,  $c$  is a  $k$ -vector,  $r$  is a scalar, and under the null hypothesis

$$c_1 \beta_1 + \dots + c_k \beta_k - r = 0.$$

For example, by setting  $c_1 = 1$ ,  $c_2 = -1$ ,  $c_3 = \dots = c_k = 0$ , and  $r = 0$ , one can test the hypothesis that  $\beta_1 = \beta_2$ .

The OLS estimator of  $\beta$  satisfies

$$\hat{\beta} \mid X \sim N\left(\beta, \sigma^2 (X^\top X)^{-1}\right). \quad (8)$$

Then,

$$\frac{c^\top \hat{\beta} - c^\top \beta}{\sqrt{\sigma^2 c^\top (X^\top X)^{-1} c}} \mid X \sim N(0, 1).$$

Therefore, under  $H_0$ ,

$$\frac{c^\top \hat{\beta} - r}{\sqrt{\sigma^2 c^\top (X^\top X)^{-1} c}} \mid X \sim N(0, 1). \quad (9)$$

Consider the  $t$ -statistic

$$\begin{aligned} T &= \frac{c^\top \hat{\beta} - r}{\sqrt{s^2 c^\top (X^\top X)^{-1} c}} \\ &= \left( \frac{c^\top \hat{\beta} - r}{\sqrt{\sigma^2 c^\top (X^\top X)^{-1} c}} \right) / \sqrt{\frac{U^\top M_X U}{\sigma^2} / (n - k)}. \end{aligned}$$

Under  $H_0$ , the result in (9) holds. Further, conditional on  $X$ ,

$$U^\top M_X U / \sigma^2 \mid X \sim \chi_{n-k}^2 \text{ and independent of } \hat{\beta}. \quad (10)$$

Therefore, under  $H_0$ ,

$$T \mid X \sim t_{n-k}.$$

Thus, the significance-level- $\alpha$  two-sided test of  $H_0 : c^\top \beta = r$  is given by

$$\text{Reject } H_0 \text{ if } |T| > t_{n-k, 1-\alpha/2}.$$

By setting the  $j$ -th element of  $c$  to  $c_j = 1$  and the rest of the elements of  $c$  equal to zero, one obtains the test discussed in the previous section:

$$\begin{aligned} H_0 : & \beta_j = r, \\ H_1 : & \beta_j \neq r. \end{aligned}$$

One rejects  $H_0$  if

$$\begin{aligned} |T| &= \left| \frac{\hat{\beta}_j - r}{\sqrt{s^2 [(X^\top X)^{-1}]_{jj}}} \right| \\ &> t_{n-k, 1-\alpha/2}, \end{aligned}$$

where  $[(X^\top X)^{-1}]_{jj}$  denotes the element  $(j, j)$  of the matrix  $(X^\top X)^{-1}$ .

## Testing multiple linear restrictions

Suppose we want to test

$$\begin{aligned} H_0 : & R\beta = r, \\ H_1 : & R\beta \neq r, \end{aligned}$$

where  $R$  is a  $q \times k$  matrix and  $r$  is a  $q$ -vector. For example,

- $R = I_k$ ,  $r = 0$ . In this case, we test that  $\beta_1 = \dots = \beta_k = 0$ .
- $R = \begin{pmatrix} 1 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \end{pmatrix}$ ,  $r = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ . In this case,  $H_0 : \beta_1 + \beta_2 = 1, \beta_3 = 0$ .

Consider the  $F$ -statistic

$$F = \left( R\hat{\beta} - r \right)^\top \left( s^2 R (X^\top X)^{-1} R^\top \right)^{-1} \left( R\hat{\beta} - r \right) / q.$$

We show next that under  $H_0$ ,

$$F \mid X \sim F_{q, n-k}. \quad (11)$$

First, it follows from (8) that

$$R\hat{\beta} \mid X \sim N \left( R\beta, \sigma^2 R (X^\top X)^{-1} R^\top \right).$$

Then, under  $H_0$ ,

$$R\hat{\beta} - r \mid X \sim N \left( 0, \sigma^2 R (X^\top X)^{-1} R^\top \right).$$

Further, by Lemma 2 in Lecture 4,

$$\left( R\hat{\beta} - r \right)^\top \left( \sigma^2 R (X^\top X)^{-1} R^\top \right)^{-1} \left( R\hat{\beta} - r \right) \sim \chi_q^2.$$

The result in (11) follows from (10) and the definition of the  $F$ -distribution. The test is given by

$$\begin{aligned} \text{Reject } H_0 \text{ if } F &= \left( R\hat{\beta} - r \right)^\top \left( s^2 R (X^\top X)^{-1} R^\top \right)^{-1} \left( R\hat{\beta} - r \right) / q \\ &> F_{q, n-k, 1-\alpha}. \end{aligned}$$

## Restricted OLS

An alternative approach to hypothesis testing is based on restricted estimation. One might consider the loss of fit resulting from choosing values other than  $\hat{\beta}$  for the regression coefficients. Consider the *restricted* LS problem

$$\min_b (Y - Xb)^\top (Y - Xb) \quad \text{s.t.} \quad Rb = r.$$

A Lagrangian function for this problem is

$$L(b, \lambda) = (Y - Xb)^\top (Y - Xb) + 2\lambda^\top (Rb - r),$$

where  $\lambda$  is a  $q$ -vector. Let  $\tilde{\beta}, \tilde{\lambda}$  be the solution, where  $\tilde{\beta}$  is the restricted LS estimator. It has to satisfy the first-order conditions

$$\frac{\partial L(\tilde{\beta}, \tilde{\lambda})}{\partial b} = 2X^\top X\tilde{\beta} - 2X^\top Y + 2R^\top \tilde{\lambda} = 0, \quad (12)$$

$$\frac{\partial L(\tilde{\beta}, \tilde{\lambda})}{\partial \lambda} = R\tilde{\beta} - r = 0. \quad (13)$$

From (12),

$$\begin{aligned} \tilde{\beta} &= (X^\top X)^{-1} (X^\top Y - R^\top \tilde{\lambda}) \\ &= \hat{\beta} - (X^\top X)^{-1} R^\top \tilde{\lambda}. \end{aligned}$$

Combining the last equation with (13),

$$\begin{aligned} r &= R\tilde{\beta} \\ &= R\hat{\beta} - R(X^\top X)^{-1} R^\top \tilde{\lambda}, \end{aligned}$$

and

$$\tilde{\lambda} = \left( R (X^\top X)^{-1} R^\top \right)^{-1} (R\hat{\beta} - r).$$

Therefore, the restricted LS estimator is given by

$$\tilde{\beta} = \hat{\beta} - (X^\top X)^{-1} R^\top \left( R (X^\top X)^{-1} R^\top \right)^{-1} (R\hat{\beta} - r).$$

Define the restricted residuals

$$\begin{aligned} \tilde{U} &= Y - X\tilde{\beta} \\ &= (Y - X\hat{\beta}) + X(X^\top X)^{-1} R^\top \left( R (X^\top X)^{-1} R^\top \right)^{-1} (R\hat{\beta} - r) \\ &= \hat{U} + X(X^\top X)^{-1} R^\top \left( R (X^\top X)^{-1} R^\top \right)^{-1} (R\hat{\beta} - r), \end{aligned}$$

where  $\hat{U}$  is the vector of unrestricted residuals. Consider the *restricted* Residual Sum of Squares:

$$\begin{aligned} RSS_r &= \tilde{U}^\top \tilde{U} \\ &= \hat{U}^\top \hat{U} + (R\hat{\beta} - r)^\top \left( R (X^\top X)^{-1} R^\top \right)^{-1} (R\hat{\beta} - r) \\ &\quad + 2\hat{U}^\top X (X^\top X)^{-1} R^\top \left( R (X^\top X)^{-1} R^\top \right)^{-1} (R\hat{\beta} - r) \\ &= RSS + (R\hat{\beta} - r)^\top \left( R (X^\top X)^{-1} R^\top \right)^{-1} (R\hat{\beta} - r), \end{aligned}$$

where  $RSS = \hat{U}^\top \hat{U}$  denotes the unrestricted Residual Sum of Squares. Since  $s^2 = \hat{U}^\top \hat{U} / (n - k)$ , the  $F$ -statistic discussed in the previous section can be written as

$$F = \frac{(RSS_r - RSS) / q}{RSS / (n - k)}. \quad (14)$$

Examples:

1. Model significance. Consider a model with an intercept

$$Y_i = \beta_1 + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + U_i.$$

Consider the null hypothesis  $H_0 : \beta_2 = \dots = \beta_k = 0$ . The restricted model is given by

$$Y_i = \beta_1 + U_i.$$

In this case, the restricted LS estimator is  $\tilde{\beta}_1 = n^{-1} \sum_{i=1}^n Y_i = \bar{Y}$ , and  $RSS_r = TSS = \sum_{i=1}^n (Y_i - \bar{Y})^2$ . Therefore,

$$\begin{aligned} F &= \frac{(TSS - RSS) / (k - 1)}{RSS / (n - k)} \\ &= \frac{ESS / (k - 1)}{RSS / (n - k)} \\ &= \frac{R^2 / (k - 1)}{(1 - R^2) / (n - k)} \\ &\sim F_{k-1, n-k} \quad \text{under } H_0. \end{aligned}$$



2. Consider the model

$$Y_i = \beta_1 + \beta_2 X_{i2} + \beta_3 X_{i3} + U_i,$$

and the null hypothesis  $H_0 : \beta_2 = \beta_3$ . The restricted model is given by

$$Y_i = \beta_1 + \beta_2 (X_{i2} + X_{i3}) + U_i.$$

Thus, to test whether  $\beta_2 = \beta_3$ , one may construct the new variable  $W_i = (X_{i2} + X_{i3})$ , compute  $RSS_r$  by taking the  $RSS$  from the regression of  $Y_i$  on a constant and  $W_i$ , compute  $RSS$  from the unrestricted regression, and construct the  $F$ -statistic according to (14).